

Abba Otieno Ndomo

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Work Authorization: F-1 | CPT available now | OPT (May 2027) | STEM OPT eligible

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Industrial & Systems Engineering – Operations Research

Atlanta, GA
Aug. 2023 – May 2027

- Honors & Awards: Dean's List, Peer Inspiration Award, 3rd Place @ OKSEF International Engineering Fair
- Coursework: Stochastic Processes, Linear & Non-Linear Optimization, Machine Learning, Data Structures & Algorithms, Database Systems, Simulations

PROFESSIONAL EXPERIENCE

Data & ML Engineer - Altura (Freelance)

Jan. 2026 – Present

Python, PySpark, Airflow, pandas, PyArrow, DuckDB, Next.js, Parquet

- Built end-to-end renovation cost estimation platform for a 101-unit multi-family property operator; automated ETL pipeline parses nested Excel pricing matrices across 13 renovation categories into 4 structured Parquet tables — estimate generation from hours to sub-5 seconds
- Architected scalable PySpark-ready Parquet schema and Airflow DAG pipeline (FileSensor → PySpark → DuckDB validation) enabling 15% faster processing and seamless multi-property expansion from single-unit MVP; Designed robust ETL architecture with PySpark-ready schema and Airflow orchestration
- Delivered full-stack Next.js estimation interface with 4-step project wizard, tier-based pricing engine (Value / Standard / Premium multipliers), and side-by-side project comparison — replacing client's manual Excel workflow entirely

Quantitative Analyst - GTSF

Jan. 2025 – May 2025

Python, Bloomberg Terminal API (blpapi), pandas, Statsmodels, NumPy, ARIMA, Monte Carlo

Atlanta, GA

- Reduced portfolio VaR by 15% validating sector-rotation strategies via ARIMA time series models and rolling correlation analysis on interest rate data across a \$500K live equity portfolio
- Conducted 10,000-iteration Monte Carlo simulations stress-testing portfolio resilience under macro shock scenarios, identifying 8% downside risk and informing strategic hedging decisions
- Built automated Python equity screener expanding ticker coverage from 50 to 120 stocks via Bloomberg Terminal API, improving portfolio diversification analysis by 30%

Data Science Intern

Jun. 2024 – Aug. 2024

Python, PostgreSQL, Statistical Analysis, System Design

Mumbai, India

- Designed production document intelligence system serving 50+ auditors; architected fault-tolerant 3-tier pipeline processing 10,000+ documents monthly with 95% classification precision (validated on held-out test, $p < 0.001$)
- Optimized end-to-end latency from 45s → 8s (82%) through algorithmic optimization ($O(n^2) \rightarrow O(n+m)$), parallel processing, and database optimization (composite indexes, connection pooling)
- Engineered statistical validation framework implementing 50+ automated checks (distributional tests, outlier detection, completeness metrics); reduced errors from 40% → 5%
- Built comprehensive testing infrastructure achieving 99% coverage; implemented A/B testing framework with bootstrapped CIs and proper multiple comparison corrections

PROJECTS

Founder - PerChance

Jan. 2026 – Present

PySpark, Airflow, PyTorch, XGBoost, Django, Postgres, Docker, Redis, React

- Built and deployed a production NBA prediction platform serving 100+ daily users — full-stack ML system with automated ingestion, model training, and live prediction serving via REST API
- Engineered PySpark pipeline processing 500,000+ NBA records with 6 custom Holt-Winters time-decay features leveraging Airflow DAGs orchestrating full ETL → retraining → serving pipeline, cutting ML prep time 90%
- Architected innovative hybrid PyTorch + XGBoost model combining deep embeddings with gradient boosting, achieving 70% accuracy across 20 metrics — 15% improvement over baseline models
- Deployed 3-tier Dockerized infrastructure (Django / Postgres / React) with Redis caching and Celery async workers; composite indexes and query optimization sustain sub-200ms API response times under live traffic

Loan Default Risk Model

Oct. 2025 – Dec. 2025

SQL, XGBoost, SHAP, Python

- Built XGBoost ensemble classifier on 50,000+ loan applications achieving 80% AUC and 82% recall; engineered 15 predictive features (payment velocity, credit utilization trends, account age) via CTE-based SQL ETL on raw credit bureau data; applied SMOTE to address 20:1 class imbalance
- Applied SHAP analysis identifying Debt-to-Income ratio as primary risk driver, creating interpretable risk tier framework that enabled stakeholders to improve loan approval decisions by 10%

Hospital Operations Database

Oct. 2025 – Dec. 2025

SQL (PostgreSQL), Python

- Designed optimized PostgreSQL schema with 3NF normalization across 15 tables, reducing data redundancy by 20% and enabling 30% faster analysis of 50,000+ patient records for hospital operations
- Developed complex SQL queries with Window functions and CTEs analyzing bed utilization; created composite indexes reducing query time (3s - 200ms)

LEADERSHIP

Housing Office Assistant

Sept. 2025 – Present

Georgia Institute of Technology Housing Department

Atlanta, GA

- Coordinated 10 monthly events serving 200+ students, achieving 95% attendance rate and 15% improvement in satisfaction scores

Chair – DEI Committee, ISyE Department

2023 – 2024

Analytics Working Group Lead, Georgia Institute of Technology

- Spearheaded data analysis of 500+ student responses, uncovering 25% satisfaction gaps and developing 2 strategic initiatives adopted into FY24 plan

TECHNICAL EXPERTISE

Finance & Quant: Bloomberg Terminal API, Time Series Analysis (ARIMA), Monte Carlo Simulation, VaR, Portfolio Optimization

Languages: Python (OOP, Pandas, NumPy), SQL (CTEs, Window Functions), Java, JavaScript, TypeScript, R, Bash

Data Engineering: PySpark, Airflow, PostgreSQL, DuckDB, Parquet, PyArrow, ETL Pipelines, 3NF / Data Modelling, EERD

Frameworks & Dev: Django, Flask, React, React Native, Next.js, Docker, Redis, Celery, REST APIs, CI/CD, PyTest, Git

Machine Learning: RAG Pipelines, NLP (Transformers), XGBoost, Scikit-Learn, SHAP, A/B Testing, Experiment Design